

Activation Function TANH :

```
import numpy as np
# Input values
x = np.array([-3, -2, -1, 0, 1, 2, 3])
# Apply Tanh activation function
output = np.tanh(x)
print("Input Values:")
print(x)
print("\nTanh Output:")
print(output)
```

... Input Values:
[-3 -2 -1 0 1 2 3]

Tanh Output:
[-0.99505475 -0.96402758 -0.76159416 0. 0.76159416 0.96402758 0.99505475]

Activation Function ReLu :

```
import numpy as np
# Input values
x = np.array([-5, -2, 0, 3, 7])
# Apply ReLU activation function
relu_output = np.maximum(0, x)
print("Input Values:")
print(x)
print("\nReLU Output:")
print(relu_output)
```

... Input Values:
[-5 -2 0 3 7]

ReLU Output:
[0 0 0 3 7]

Activation Function Sigmoid :

```
import numpy as np
# Input values
x = np.array([-5, -2, 0, 2, 5])
# Apply Sigmoid activation function
sigmoid_output = 1 / (1 + np.exp(-x))
print("Input Values:")
print(x)
print("\nSigmoid Output:")
print(sigmoid_output)
```

```
... Input Values:
[-5 -2  0  2  5]

Sigmoid Output:
[0.00669285 0.11920292 0.5          0.88079708 0.99330715]
```

Activation Function SOFTMAX :

```
import numpy as np
marks = np.array([85, 70, 60])
softmax = np.exp(marks) / np.sum(np.exp(marks))
print("Scores :", marks)
print("Softmax:", softmax)
```

```
... Scores : [85 70 60]
Softmax: [9.99999694e-01 3.05902227e-07 1.38879396e-11]
```